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Domain :- AI, Deep Learning & DWH Mining

SDG :-

SDG 3 (Good Health) | **SDG 10** (Reduced Inequalities) | **SDG 9** (Innovation)

Title of the Project : **CogniSense** - Multimodal Digital Biomarkers for Pre-clinical Dementia Screening

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Abstract

Alzheimer's Disease and related dementias represent a growing global health crisis, with India facing a significant diagnostic gap due to high costs and a lack of specialized neurologists. Current clinical diagnoses, such as MRIs and PET scans, typically occur at the "Symptomatic Stage", by which point irreversible neuronal damage has already occurred.

This project proposes a smartphone-based "Digital Triage" system designed to detect pre-clinical signs of Alzheimer's 5–7 years before visible symptoms appear. By leveraging Deep Learning and Data Warehouse (DWH) Mining, the system analyzes a multimodal set of digital biomarkers: acoustic speech patterns, oculomotor (eye) movements, and visuospatial drawing metrics. Unlike existing solutions that require specialized hardware, our framework utilizes standard smartphone sensors, making it an ideal tool for General Physicians and Family Doctors to use during routine geriatric checkups.

The inclusion of DWH Mining allows for longitudinal tracking of a patient's "Brain Velocity", identifying subtle rates of decline over time. This non-invasive approach aims to bridge the diagnostic gap, providing an affordable, language-agnostic screening tool for the Indian middle-class and rural populations.

Introduction

Dementia is not a normal part of aging, yet it is often misdiagnosed or ignored until it reaches advanced stages. In the Indian context, the challenge is three-fold : an Accessibility Crisis, where the neurologist-to-patient ratio is extremely low; a Cost Barrier, where standard screenings like PET scans cost upwards of ₹15,000; and a Diagnostic Gap, where medical intervention only begins after significant cognitive loss.

The biological onset of Alzheimer's begins nearly two decades before the first memory lapse. During this "pre-clinical" phase, subtle physiological changes occur in how a person speaks, moves their eyes, and executes fine motor tasks. While these changes are often invisible to the naked eye, they leave "Digital Fingerprints" that AI can identify.

CogniSense shifts the focus from reactive hospital-grade testing to proactive community-based screening. Our project focuses on three primary pillars :

- Speech & Acoustic Analysis : Capturing micro-hesitations and pitch variations that signal early cognitive load.
- Oculomotor Metrics : Using standard front-facing cameras to track eye-jitters and response times to visual stimuli.
- Visuospatial Data : Digitizing traditional tests like the "Clock Drawing Test" to analyze stroke velocity and tremors via Vision Transformers.

A key innovation of this project is its integration into the Primary Healthcare (PHC) ecosystem. By equipping family doctors with an easy-to-use digital tool, we move the point of detection from the specialist's office to the local clinic. Furthermore, by utilizing Data Warehouse Mining, we don't just provide a "Yes/No" result; we track a patient's cognitive trajectory over months and years. This longitudinal data is crucial for identifying the "Rate of Decline", which is often a more accurate predictor of Alzheimer's than a single snapshot test.

Problem Statement

Background & Context

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder and the most common cause of dementia, accounting for 60–70% of all dementia cases globally. According to the World Health Organization (WHO), over 55 million people worldwide currently live with dementia, a figure projected to reach 139 million by 2050. In India alone, the Dementia India Report estimates approximately 8.8 million individuals are currently affected, with this number expected to double every 20 years due to a rapidly aging population.

The fundamental tragedy of Alzheimer's Disease is not merely its prevalence, but the timing of its detection. The pathological cascade of AD — characterized by amyloid-beta plaque accumulation and tau protein tangles — begins 15 to 20 years before the first observable clinical symptom. This extended pre-clinical window represents a critical, largely untapped opportunity for intervention. Yet, the current healthcare infrastructure in India fails to exploit this window entirely, resulting in a pattern where diagnosis consistently occurs at a stage when neuronal damage is already severe and irreversible.

The Three-Dimensional Problem :

The failure to achieve early Alzheimer's detection in India is not attributable to a single cause. It is the result of three deeply interconnected crises that compound one another :

1. The Diagnostic Gap — Detection That Arrives Too Late

   Current gold-standard diagnostic tools for Alzheimer's include :

   - Positron Emission Tomography (PET) Scans — detect amyloid-beta deposits in the brain
   - Cerebrospinal Fluid (CSF) Analysis — measures tau and amyloid protein biomarkers via lumbar puncture
   - Structural MRI — identifies hippocampal atrophy and cortical thinning
   - Neuropsychological Battery Tests — cognitive assessments like MMSE, MoCA

   The critical limitation of these tools is that they are deployed reactively — only after a patient or family member reports visible cognitive decline, such as memory lapses, disorientation, or personality changes. By this stage, clinical studies confirm that 40 to 60 percent of neurons in the hippocampus and entorhinal cortex have already been permanently lost.

   Furthermore, India lacks any structured mass cognitive screening protocol for its

aging population. Unlike cardiovascular disease screening (BP checks) or diabetic screening (HbA1c tests) which are routinely administered, cognitive health receives zero attention at the primary healthcare level until symptoms become undeniable.

2. The Accessibility Crisis — A Specialist Deficit

Metric	India	WHO Recommendation
Neurologists per 100,000 population	**0.07**	**1.0**
Total practicing neurologists	**~2,500**	**~14,000 (needed)**
Neurologists in rural India	**<5% of total**	—
Average wait time for neurology appointment	**4–12 weeks**	—

Sources : Indian Academy of Neurology, 2022; WHO Neurology Atlas, 2023

This means that for a patient in rural Maharashtra, Uttar Pradesh, or Bihar, accessing a neurologist for cognitive evaluation is not merely inconvenient — it is practically impossible. The specialist ecosystem that exists is concentrated in Tier-1 cities, creating a geographic healthcare inequality that systematically excludes the populations most vulnerable to underdiagnosis.

3. The Cost Barrier — Screening That Excludes the Many

   The financial burden of Alzheimer's diagnosis in India places it entirely out of reach for the middle class and rural populations :

Diagnostic Procedure	Approximate Cost in India	Availability
FDG-PET Scan	₹15,000 – ₹25,000	Metro cities only
Amyloid-PET Scan	₹40,000 – ₹80,000	3–4 cities nationally
CSF Biomarker Analysis	₹8,000 – ₹15,000	Limited labs
Brain MRI (3T)	₹5,000 – ₹12,000	Tier-1/2 cities
Full Neuropsychological Battery	₹3,000 – ₹8,000	Specialist centers only

Formal PS :

"There exists no affordable, non-invasive, hardware-independent, and longitudinally-aware screening tool that enables General Physicians and Family Doctors in India to objectively assess pre-clinical cognitive decline in elderly patients during routine checkups — resulting in a systemic failure to detect Alzheimer's Disease before irreversible neuronal damage has occurred.

The absence of such a tool creates a critical diagnostic gap where early intervention opportunities are consistently missed, disease-modifying therapies cannot be administered at the optimal therapeutic window, and the caregiving burden on families and the healthcare system escalates unnecessarily."

Proposed Solution

Solution Philosophy — "Step 0 Digital Triage"

CogniSense is not designed to replace neurologists. It is designed to dramatically reduce the number of people who never reach one.

The solution philosophy is modeled on the concept of "Step 0 Triage" — inserting a low-cost, high-sensitivity screening layer into the existing primary healthcare touchpoint (the GP/family doctor visit) that currently has zero cognitive screening capability. A positive flag from CogniSense does not constitute a diagnosis; it constitutes a clinically-informed referral decision, replacing the current practice of either ignoring cognitive concerns or making expensive specialist referrals based purely on subjective observation.

Proposed Solution : System Overview

CogniSense is a tablet/smartphone-based clinical decision support application administered by a General Physician during a routine geriatric checkup. The complete assessment takes 5–8 minutes and requires no specialized hardware beyond a standard Android tablet with a front-facing camera.

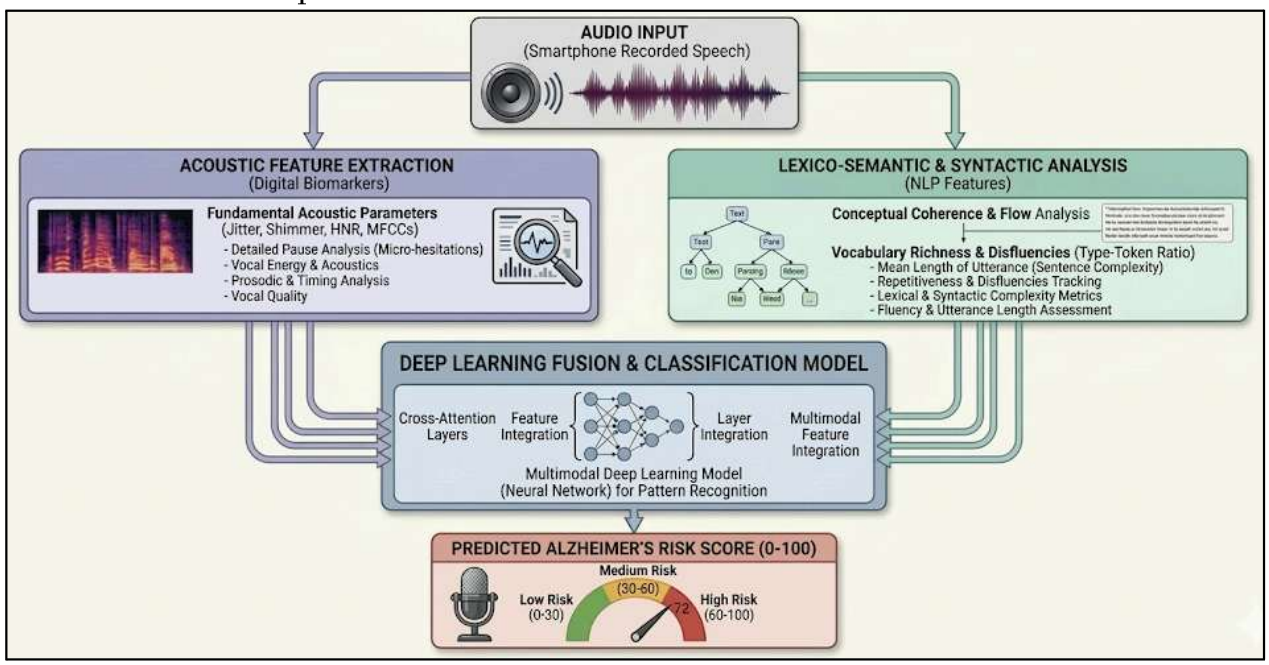
The system captures three categories of digital biomarkers through three structured tasks, processes them through independent deep learning modules, fuses the outputs through a Cross-Attention Transformer, and generates a Cognitive Risk Report stored in a longitudinal Data Warehouse.

Module 1 — Speech & Acoustic Analysis

Clinical Basis : Research by Konig et al. (2015) and Fraser et al. (2016) demonstrates that pre-clinical AD patients exhibit measurable changes in spontaneous speech including increased pause frequency, reduced semantic density, simplified syntactic structure, and altered fundamental frequency (pitch) — years before memory complaints arise.

Task Administered : The Cookie Theft Picture Description Task (from the Boston Diagnostic Aphasia Examination) — a standardized stimulus image where the patient is asked to verbally describe everything they see for 60 seconds. This is the same task used in the DementiaBank Pitt Corpus, ensuring direct alignment with available training data.

Feature Extraction Pipeline :



Key Innovation — Language Agnosticism : By prioritizing acoustic features (pauses, pitch, energy) over lexical content, the system remains functional even when the patient speaks in Hindi, Marathi, Tamil, or other regional languages. Wav2Vec 2.0 multilingual variants (XLS-R) support 128 languages, enabling transcript-based features for major Indian languages as well.

Module 2 — Oculomotor Analysis (Eye Tracking)

Clinical Basis : Oculomotor dysfunction is one of the earliest measurable biomarkers of Alzheimer's Disease. Studies by Shafiq-Antonacci et al. and Crawford et al. demonstrate that AD patients exhibit significantly elevated antisaccade error rates, prolonged saccadic latency, and irregular smooth pursuit patterns — all detectable years before cognitive symptoms appear. The anterior cingulate cortex and frontal eye fields, which govern voluntary eye movement control, are among the earliest regions affected by tau pathology.

Task Administered :

TASK A — Prosaccade Task (2 minutes)

- A dot appears randomly on the LEFT or RIGHT of the screen.
- Patient must look TOWARD the dot as quickly as possible.
- Measures : Reaction time, accuracy (baseline calibration)

TASK B — Antisaccade Task (3 minutes) ← PRIMARY BIOMARKER

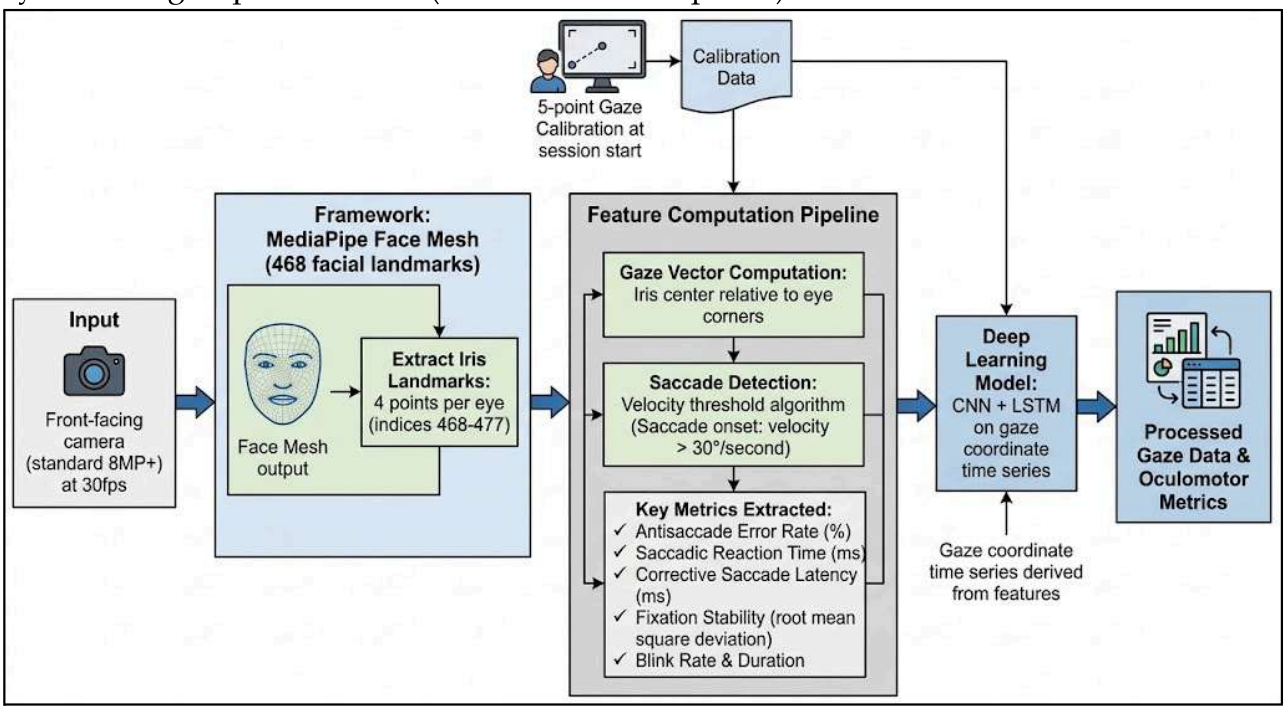
- A dot appears randomly on one side.

- Patient must look to the OPPOSITE side (inhibitory control).
- Measures : Error rate, correction latency, anticipatory saccades
- This task directly stresses the prefrontal cortex — first region degraded in pre-clinical AD.

TASK C — Smooth Pursuit Task (1 minute)

- A dot moves in a sinusoidal horizontal path.
- Patient must follow it with their eyes.
- Measures : Pursuit gain, catch-up saccade frequency

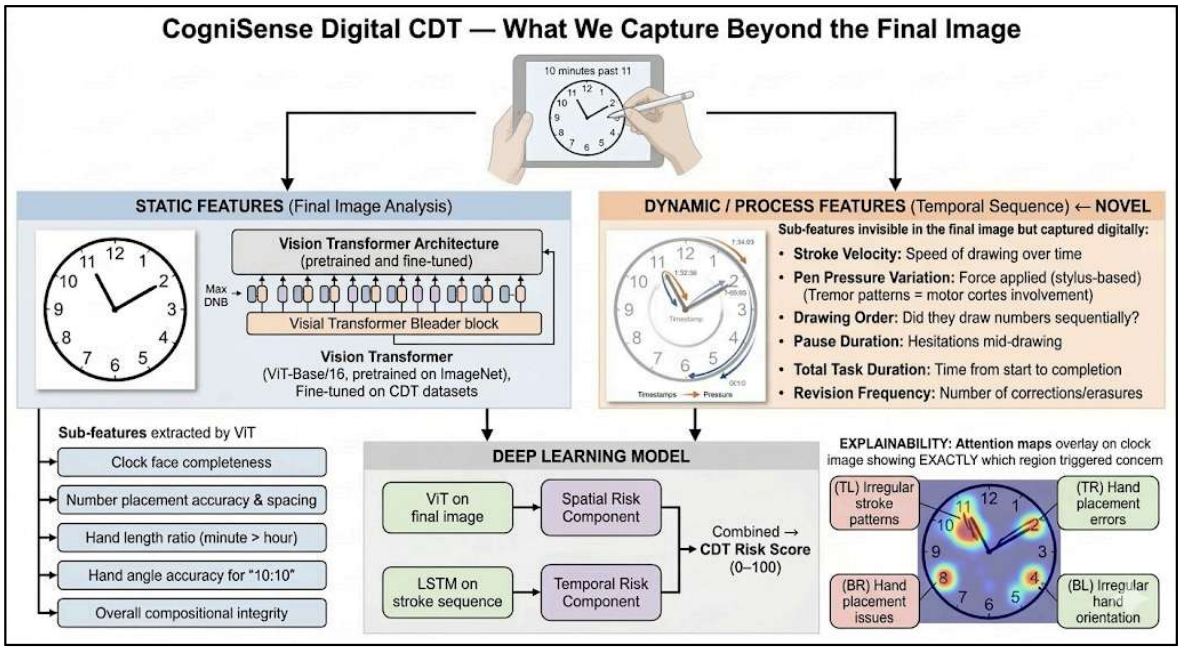
Eye Tracking Implementation (No Hardware Required):



Key Innovation: Existing oculomotor research relies on hardware eye-trackers (Tobii, EyeLink) costing ₹5–15 lakhs. CogniSense achieves comparable metric extraction using only the front-facing camera via MediaPipe, making this clinically-informed oculomotor assessment available at zero additional hardware cost.

Module 3 — Visuospatial Analysis (Digital Clock Drawing Test)

Clinical Basis : The Clock Drawing Test (CDT) is one of the most widely validated neuropsychological screening tools, sensitive to visuospatial, executive function, and motor planning deficits associated with AD. However, traditional CDT scoring is subjective (inter-rater variability is a documented problem), static (captures only the final image), and insensitive to process features (how the clock was drawn is diagnostically as important as what the final clock looks like).

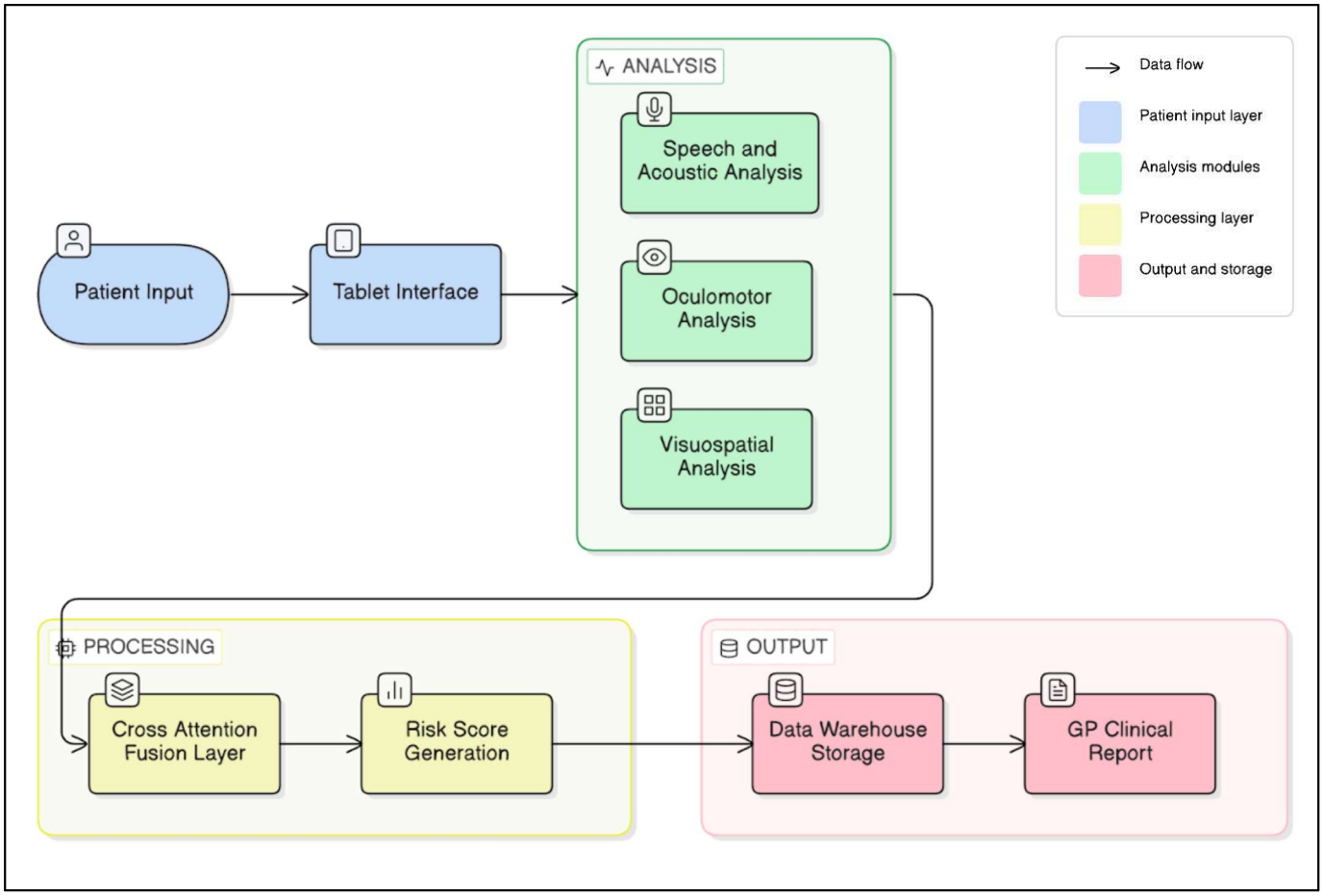


Solution Summary Table

Dimension	Current State	CogniSense Solution
Detection Stage	Symptomatic (40-60% neurons lost)	Pre-clinical / MCI (intervention possible)
Cost per Screening	₹15,000-₹80,000	₹5-₹15
Hardware Required	PET scanner / MRI / EyeTracker	Standard Android tablet
Time Required	2-4 hours (hospital visit)	5-8 minutes (GP clinic)
Who Administers	Neurologist / Radiologist	General Physician
Result Type	Single-point diagnosis	Longitudinal trajectory + Risk score
Explainability	Raw scan / specialist interpretation	Plain-language GP-readable report
Language Barrier	English-centric tests	Acoustic features = language agnostic
Tracking	Repeated expensive scans	Continuous DWH-based Brain Velocity
Availability	Metro cities only	Any clinic with a tablet

Methodology

The CogniSense system operates as a "Step 0 Digital Triage" tool, designed to be seamlessly integrated into primary healthcare settings. The evaluation is conducted through a structured workflow :



- **Data Acquisition :** The general physician administers three structured tasks via a standard Android tablet during a routine 5-8 minute geriatric checkup.

- **Speech and Acoustic Processing :** The patient performs the Cookie Theft Picture Description Task while the device's microphone records the audio. The system extracts fundamental acoustic parameters, micro-hesitations, and NLP-based lexico-semantic features.

- **Oculomotor Tracking :** Utilizing the tablet's front-facing camera, the system tracks eye movements during prosaccade, antisaccade, and smooth pursuit tasks. It maps 468 facial landmarks to isolate iris movements and compute saccadic metrics without specialized hardware.

- **Visuospatial Processing :** The patient completes a digitized Clock Drawing Test on the tablet interface. The system captures both static final-image features and dynamic temporal sequences, such as stroke velocity and pen pressure.

- **Multimodal Fusion :** Data from all three testing modules is fed into independent deep learning pipelines, including a CNN+LSTM for gaze data and a Vision Transformer for the visuospatial test. A cross-attention model integrates these features to generate a comprehensive predicted Alzheimer's risk score.

- **Longitudinal Tracking :** The generated risk reports are stored in a data warehouse (DWH) to track the patient's cognitive trajectory and identify subtle rates of decline over months and years.

Hardware, Software and tools Requirements

Category	Requirement	Project Application
Hardware	Android tablet or smartphone	Serves as the primary testing interface for the patient and physician, requiring no specialized clinical equipment.
Hardware	Front-facing camera (8MP+ at 30fps)	Captures real-time video input necessary for tracking oculomotor saccades and iris landmarks.
Hardware	Standard device microphone	Records the patient's spontaneous speech during the descriptive audio task.
Software	MediaPipe framework	Provides the face mesh technology to extract exact facial and iris coordinates.
Software	Deep learning models (CNN, LSTM, ViT)	Processes complex time-series gaze data and analyzes spatial clock-drawing sequences.
Software	Wav2Vec 2.0 (XLS-R)	Enables extraction of transcript-based and acoustic features across multiple regional languages.
Database	Data warehouse (DWH) infrastructure	Facilitates the continuous storage and mining of patient results to track longitudinal brain velocity.

Proposed Evaluation Measures

The performance of the CogniSense system will be evaluated across three dimensions: modular accuracy, fusion model performance, and clinical utility.

Module-Level Evaluation

Module	Primary Metric	Target Benchmark
Speech & Acoustic Analysis	AUC-ROC, F1-Score	AUC > 0.82 (vs. DementiaBank baseline)
Oculomotor Analysis	Antisaccade Error Rate Correlation	r > 0.75 with clinical ground truth
Visuospatial (CDT)	Classification Accuracy (ViT)	Accuracy > 85% on ADNI-CDT subset
Multimodal Fusion Model	AUC-ROC, Sensitivity, Specificity	AUC > 0.88, Sensitivity > 80%
Longitudinal Brain Velocity	Rate-of-Decline Tracking (MAE)	MAE < 0.05 per 6-month interval

Clinical Performance Metrics

The system will be evaluated using the following standard clinical screening benchmarks:

- **Sensitivity (Recall) :** Proportion of true pre-clinical AD cases correctly flagged. Target ≥ 80%, prioritizing reduction of false negatives.
- **Specificity :** Proportion of healthy individuals correctly identified as low-risk. Target ≥ 75%.
- **Positive Predictive Value (PPV) :** Relevance of a positive CogniSense flag. To be compared against current GP-only subjective assessment baseline.
- **Area Under the ROC Curve (AUC-ROC) :** Primary scalar metric for overall discriminative performance across all classification thresholds.

Explainability and Usability Evaluation

- **SHAP / Attention Map Validity :** Clinical experts will assess whether the highlighted biomarker features align with neurologically established indicators of early AD.
- **Physician Usability Score :** System Usability Scale (SUS) questionnaire

administered to General Physicians in a simulated pilot setting.

- **Report Clarity Score :** Structured feedback from target users (GPs and family doctors) on the comprehensibility of the generated Cognitive Risk Report.

Longitudinal Validation

The DWH-based longitudinal tracking module will be evaluated by simulating repeated assessments (minimum 3 time-points, 6 months apart) using publicly available longitudinal datasets (ADNI, NACC). The stability of the "Brain Velocity" score across assessments will be measured using intra-class correlation coefficients (ICC).

Conclusion

CogniSense addresses a critical and well-documented gap in India's healthcare ecosystem: the near-complete absence of an affordable, non-invasive, and longitudinally-aware cognitive screening tool at the primary healthcare level. The current state of Alzheimer's detection in India is defined by three compounding failures — a diagnostic gap that consistently results in late-stage detection, an accessibility crisis driven by a severe neurologist deficit, and a cost barrier that renders gold-standard diagnostics unavailable to the majority of the population.

This project proposes a practical and technically grounded solution. By leveraging multimodal digital biomarkers — spontaneous speech acoustics, oculomotor saccadic dynamics, and visuospatial drawing metrics — captured entirely through standard smartphone hardware, CogniSense enables objective, data-driven cognitive assessment in any general physician's clinic. The integration of a Cross-Attention Transformer for multimodal fusion, Explainable AI for clinician-readable output, and a Data Warehouse for longitudinal "Brain Velocity" tracking positions CogniSense not merely as a diagnostic aid, but as a continuous cognitive health monitoring system.

The key innovations of this project are its language-agnosticism (enabling use across India's multilingual population), its zero-additional-hardware deployment model, and its positioning as a "Step 0 Triage" tool — one that does not replace neurological expertise, but ensures that significantly more patients receive timely, evidence-based referrals. If even a fraction of the estimated 8.8 million affected individuals in India are identified one to two years earlier than current practice allows, the reduction in irreversible neuronal loss, caregiver burden, and downstream healthcare costs would be substantial.

CogniSense represents the convergence of Deep Learning, Data Warehouse Mining, and public health necessity — a technically rigorous, socially relevant, and clinically feasible contribution toward SDG 3 (Good Health and Well-Being), SDG 9 (Industry, Innovation and Infrastructure), and SDG 10 (Reduced Inequalities).

References

1. Alzheimer's Disease International. (2023). *World Alzheimer Report 2023: Reducing dementia risk.* ADI, London.

2. World Health Organization. (2021). *Global status report on the public health response to dementia.* WHO Press, Geneva.

3. Shaji, K. S., Jotheeswaran, A. T., Girish, N., Bharath, S., Dias, A., Pattabiraman, M., & Prince, M. (2010). *The Dementia India Report 2010: Prevalence, impact, costs and services for dementia.* Alzheimer's and Related Disorders Society of India (ARDSI), New Delhi.

4. Fraser, K. C., Meltzer, J. A., & Rudzicz, F. (2016). Linguistic features identify Alzheimer's disease in narrative speech. *Journal of Alzheimer's Disease*, 49(2), 407–422.

5. König, A., Satt, A., Sorin, A., Hoory, R., Toledo-Ronen, O., Derreumaux, A., ... & Robert, P. H. (2015). Automatic speech analysis for the assessment of patients with predementia and Alzheimer's disease. *Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring*, 1(1), 112–124.

6. Shafiq-Antonacci, R., Maruff, P., Masters, C., & Currie, J. (2003). Spectrum of saccade system function in Alzheimer disease. *Archives of Neurology*, 60(9), 1272–1278.

7. Crawford, T. J., Higham, S., Renvoize, T., Patel, J., Dale, M., Suriya, A., & Tetley, S. (2005). Inhibitory control of saccadic eye movements and cognitive impairment in Alzheimer's disease. *Biological Psychiatry*, 57(9), 1052–1060.

8. Cummings, J. L. (1993). Frontal-subcortical circuits and human behavior. *Archives of Neurology*, 50(8), 873–880.

9. Balagopalan, A., Eyre, B., Robin, J., Rudzicz, F., & Novikova, J. (2021). To BERT or not to BERT: Comparing speech and language-based approaches for Alzheimer's disease detection. *Interspeech 2021*, 3299–3303.

10. Petti, U., Baker, S., & Korhonen, A. (2020). A systematic literature review of automatic Alzheimer's disease detection from speech and language. *Journal of the American Medical Informatics Association*, 27(11), 1784–1797.

11. Baevski, A., Zhou, Y., Mohamed, A., & Auli, M. (2020). wav2vec 2.0: A framework for self-supervised learning of speech representations. *Advances in Neural Information Processing Systems (NeurIPS)*, 33, 12449–12460.

12. Cimpoi, M., Maji, S., Kokkinos, I., & Vedaldi, A. (2015). Deep filter banks for texture recognition, description, and segmentation. *International Journal of Computer Vision*, 118(1), 65–94.

13. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., ... & Houlsby, N. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. *International Conference on Learning Representations (ICLR)*.

14. Luong, M. T., Pham, H., & Manning, C. D. (2015). Effective approaches to attention-based neural machine translation. *Proceedings of EMNLP 2015*, 1412–1421.

15. Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems (NeurIPS)*, 30, 4765–4774.

16. Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., ... & Chintala, S. (2019). PyTorch: An imperative style, high-performance deep learning library. *NeurIPS*, 32.

17. Lugaresi, C., Tang, J., Nash, H., McClanahan, C., Uboweja, E., Hays, M., ... & Grundmann, M. (2019). MediaPipe: A framework for building perception pipelines. *arXiv preprint arXiv:1906.08172*.

18. Indian Academy of Neurology. (2022). *Consensus statement on dementia from the Indian Academy of Neurology*. Annals of Indian Academy of Neurology, 25(Suppl 2).

19. Morris, J. C., Weintraub, S., Chui, H. C., Cummings, J., DeCarli, C., Ferris, S., ... & Kukull, W. A. (2006). The Uniform Data Set (UDS): Clinical and cognitive variables and descriptors for Alzheimer Disease centers. *Alzheimer Disease & Associated Disorders*, 20(4), 210–216.

20. Petersen, R. C., Aisen, P. S., Beckett, L. A., Donohue, M. C., Gamst, A. C., Harvey, D. J., ... & Weiner, M. W. (2010). Alzheimer's Disease Neuroimaging Initiative (ADNI): Clinical characterization. *Neurology*, 74(3), 201–209.